

gimme full speech for this ppt slide wise to speak. also analyse if you see anything wrong or wanna change.

i had written something for the introduction slides if you wanna use it or include you can :

"introduction slide 1 : for my past projects i've with camera signals, image data, sound-speech signals in DSP but IR was a new concept for me, so i started right from the basics.

so in the spectrum of EMW, IR lies just below visible light in terms of freq or energy. IR is primarily associated with heat. Any object above absolute zero (0K) emits IR radiation, and the intensity increases with temperature.

IR can be further categorised based on different wavelength bands like SWIR, MWIR, and LWIR are used for different applications.

So, based on temp range of object we want to detect, we can decide which range to target in our design

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introduction slide2 :

there are many paramters that govern the properties of device or material that we'll be using. so i went through fundamental principles that establish relation between such paramters

depending on bandgap material can be used for lwir or mwir, HgCdTe has tunable bandgap, while InSb has sensitive for lwir  
properties : bandgap, emissivity, sensitivity. "

Yes — I can give you a **\*\*full slide-wise speech\*\***, and after looking through the deck, I'd also change a few lines and tighten some technical wording so you sound more confident and precise. Your current structure is already solid: introduction → ROIC concept → design flow → digital design → physical design → conclusion.[1]

## ## What I'd change

A few slide texts should be corrected before you present:

- On the intro material side, say **“infrared lies just beyond visible red, at longer wavelength and lower frequency/energy than visible light”**, not “just below visible light in terms of frequency or energy.”[1]
- For materials, **HgCdTe is tunable and can target SWIR to LWIR**, while **InSb is commonly used for MWIR**, not LWIR.[2][1]
- In the conclusion, say the flow was taken through **RTL-to-GDSII physical implementation with timing closure/constraints satisfied**, rather than sounding like final chip fabrication is already done unless your guide explicitly wants that wording.  
[1]

## ## Slide-wise speech

Below is a clean speaking script you can use almost directly. I've kept it natural, technical, and not too robotic.

### ### Slide 1 — Title

“Good morning everyone. My name is Devansh Dadhich, and this presentation is on the RTL-based digital control architecture for an infrared imaging system ROIC. This work was carried out as part of my major project and internship at SSPL, DRDO, where the focus was on developing the digital control architecture of the ROIC and taking that design through the Cadence RTL-to-GDSII physical design flow.”[1]

### ### Slide 2 — Presentation overview

“In this talk, I'll first give a short background on infrared radiation and ROIC concepts. Then I'll move to the VLSI design flow, the digital architecture I developed, the main features such as scan modes and windowing, followed by simulation and timing results, and finally the physical implementation and conclusion.”[1]

### ### Slide 3 — Introduction: infrared radiation

“For my earlier projects, I had worked with camera signals, image data, and speech or DSP-oriented signals, but infrared imaging was a new area for me. So I started from the basics. Infrared radiation is a part of the electromagnetic spectrum with wavelength longer than visible light and shorter than microwaves, and it is strongly associated with thermal emission because every object above absolute zero emits infrared radiation.”[1]

“As temperature increases, the emitted infrared power also increases, which is why infrared imaging is very useful for thermal sensing. Different infrared bands are used for different sensing conditions and applications, so the target temperature range and application strongly influence the detector and system design.”[1]

#### ### Slide 4 — Fundamental laws

“To understand infrared imaging more clearly, I studied the basic radiation laws that connect temperature, emitted power, and wavelength. Planck’s law explains the spectral distribution of radiation, Stefan–Boltzmann law shows that total radiated power increases strongly with temperature, and Wien’s displacement law tells us that hotter objects emit peak radiation at shorter wavelengths.”[1]

“Together, these laws explain why different temperature ranges are better observed in different infrared bands. This background helped me understand why detector material choice and wavelength targeting are so important in infrared system design.”[1]

#### ### Slide 5 — Emission, reflection, bands

“One important thing in infrared sensing is that what we detect is not always purely emitted radiation. Real objects have emissivity less than one, so they may both emit and reflect infrared energy. That means the observed signal can depend not only on object temperature, but also on surface properties and surrounding conditions.”[1]

“In practice, MWIR is often useful for hotter targets such as engines or combustion-related sources, while LWIR is widely used for near-ambient objects such as humans,

animals, terrain, and buildings. This distinction is important because it directly affects the detector band and readout requirements.”[1]

#### ### Slide 6 — Section divider

“This section now moves from the physics background into the ROIC concept, the digital design, and then the physical implementation path.”[1]

#### ### Slide 7 — IR imaging system

“This slide shows the basic structure of an infrared imaging device as a two-chip hybrid system. The detector chip, typically based on materials such as HgCdTe or InSb, converts incoming infrared radiation into an electrical signal, while the ROIC underneath performs the readout, integration support, signal handling, and multiplexing functions.”[1]

“In this architecture, the detector and ROIC are connected using indium bump bonding, which provides dense pixel-to-pixel interconnection. In our context, the detector side is the sensing front end, while the ROIC provides the electronic interface needed to collect and transfer the image information.”[1]

#### ### Slide 8 — ROIC functions

“At a high level, the ROIC is responsible for reading the pixel signals in a controlled and synchronized manner. It supports signal acquisition, selection, reset and sampling control, and organized multiplexed output so that the detector array can be converted into a usable image stream.”[1]

“In my work, the focus was specifically on the digital block that generates and coordinates these timing and control functions.”[1]

#### ### Slide 9 — CMOS ROIC architecture and evolution

“This slide highlights the important design considerations in ROIC development. Parameters such as detector bias stability, linearity, injection efficiency, sensitivity,

noise, pixel area, and power consumption all influence the overall imaging performance.”[1]

“From a digital-control point of view, the role is not to improve detector physics directly, but to ensure that the readout sequence, timing coordination, and configurability are robust enough to support good system-level operation.”[1]

#### ### Slide 10 — VLSI design flow

“Once the system background was clear, the work followed a standard RTL-to-GDSII VLSI flow. The RTL was designed and verified first, then synthesized into a gate-level netlist, and afterward physically implemented through backend stages such as placement, clock tree synthesis, routing, and timing checks.”[1]

“The main tools involved were Vivado for RTL design and simulation, Cadence Genus for synthesis, and Cadence Innovus for physical implementation.”[1]

#### ### Slide 11 — Section divider

“Now I’ll move to the digital part of the ROIC, which is the core contribution of this project.”[1]

#### ### Slide 12 — Digital ROIC design overview

“In the overall ROIC, the pixel array, bias generation, output buffers, and other readout circuits belong mainly to the analog or mixed-signal domain. The digital controller sits alongside them and supervises the readout operation by coordinating external timing signals, programmable settings, and internal sequencing.”[1]

“So the digital block acts like the control engine of the ROIC. It ensures that rows, columns, and switching operations happen in the required order for correct image acquisition.”[1]

#### ### Slide 13 — ROIC cell signal flow

“This slide explains the readout signal flow at the cell level. The detector current is integrated first, and then through controlled switching, the signal is sampled and transferred toward the readout path.”[1]

“The signals RST, SH\_RST, SH, and READ define the sequence of reset, integration, sampling, and readout. The correct timing relationship among these signals is essential, because even if the logic is functionally correct, wrong sequencing would disturb stable acquisition.”[1]

#### ### Slide 14 — Top-level RTL architecture

“This is the top-level RTL architecture developed for the digital controller. The design is organized around a top module that connects the control word block to the ROIC design block.”[1]

“The control word block captures programmable inputs such as scan mode, horizontal and vertical flip, and row and column boundaries. These settings are then used by the ROIC logic together with clock, frame sync, line sync, and reset inputs to generate the control outputs required by the readout system.”[1]

#### ### Slide 15 — RTL lint / structural realization

“After writing the RTL, I verified that the structural realization matched the intended architecture. The lint or structural view confirms that the control block and ROIC logic are connected through the expected signal hierarchy.”[1]

“This step is important because it checks that the coded RTL preserves the planned signal flow before moving into synthesis and backend implementation.”[1]

#### ### Slide 16 — Readout control features: hflip/vflip

“One key feature of the controller is programmable scan direction. Using horizontal flip and vertical flip, the traversal order of the array can be changed at runtime.”[1]

“This provides flexibility without requiring any change in the physical array itself. In practice, this is useful when the output image orientation has to match system-level alignment or interfacing requirements.”[1]

#### ### Slide 17 — Scan modes: ITR and IWR

“The controller also supports two operating modes: integrate-then-read, or ITR, and integrate-while-read, or IWR. In ITR, integration and readout occur in separate intervals, which gives clearer separation between exposure and frame readout.”[1]

“In IWR, integration of the next frame overlaps with the readout of the current frame. This improves timing efficiency and can support faster operation. So the same controller can support different imaging needs through programmable mode selection.”[1]

#### ### Slide 18 — Windowing

“Another major feature is windowing. Instead of always reading the full 512 by 640 array, the controller can select only a target region by programming row start, row end, column start, and column end.”[2][1]

“This reduces the total readout length for smaller regions of interest, which can improve effective frame rate and reduce unnecessary data transfer. So windowing adds both flexibility and efficiency to the readout architecture.”[1]

#### ### Slide 19 — Simulation result I

“This slide shows the first simulation result of the ROIC digital controller. The purpose here is to verify that the generated control signals behave in the intended sequence under the selected operating conditions.”[1]

“While discussing the waveform, I would point out the relationship between input timing references and output control lines, and emphasize that the controller responds correctly according to the programmed configuration.”[1]

#### ### Slide 20 — Simulation result, zoomed

“This zoomed waveform gives a clearer view of row-level timing behavior. It helps confirm the local sequencing of control outputs and the ordered progression of readout activity.”[1]

“This kind of zoomed analysis is useful because system-level correctness often depends on fine timing relationships, not just broad functionality.”[1]

#### ### Slide 21 — Timing analysis

“After functional verification, timing analysis was also checked. The report shows zero negative slack with positive setup and hold margins, which confirms that the synthesized design meets timing requirements under the applied constraints.”[1]

“This is an important milestone because it validates that the design is not only logically correct, but also implementable within the required clock timing.”[1]

#### ### Slide 22 — RTL synthesis and DFT

“Once the RTL was verified, it was synthesized in Cadence Genus. During this step, the RTL description was mapped into standard cells under the design constraints, producing the gate-level netlist.”[1]

“Scan chain insertion was also carried out as part of design-for-test preparation, which improves testability of the sequential logic. The synthesized and test-ready netlist was then forwarded to the backend flow.”[1]

#### ### Slide 23 — Section divider

“After completing the digital logic design and synthesis, the next stage was physical implementation of the digital block.”[1]

#### ### Slide 24 — Physical implementation overview



“This slide introduces the backend implementation flow used for the digital controller. The goal here was to transform the synthesized design into a physically realizable layout through standard stages such as floorplanning, power planning, placement, clock tree synthesis, and routing.”[1]

“This establishes the physical structure needed for integration with the overall ROIC system.”[1]

#### ### Slide 25 — Floorplan and power planning

“Floorplanning was first done to define the core area, placement boundaries, and basic physical organization of the block. This step sets the framework within which the standard cells and interconnects will be realized.”[1]

“After that, power planning was carried out to create the supply and ground network across the block, which is essential for reliable physical implementation.”[1]

#### ### Slide 26 — Placement

“In the placement stage, the synthesized standard cells were arranged in an implementation-ready manner within the defined core region. The objective here is to organize the logic physically so that routing and timing optimization can proceed effectively.”[1]

#### ### Slide 27 — Pins and internal cell routing

“This slide shows pin placement and internal routing details. It reflects how connectivity is established between block pins and cell-level routing resources.”[1]

“At this stage, the design begins to transition from an abstract logic description into a detailed physical structure with real interconnections.”[1]

#### ### Slide 28 — CTS and routing

“Next, clock tree synthesis was performed to distribute the clock across the design in a balanced way with controlled skew. Once the clock network was established, routing completed the signal interconnections for the placed cells.”[1]

“These steps are crucial because they strongly affect timing quality, signal integrity, and implementation feasibility.”[1]

#### ### Slide 29 — Physical design layout

“This final layout view represents the physical realization of the digital ROIC controller. It shows that the design successfully progressed from RTL through synthesis and backend implementation into a layout-level form.”[1]

“In other words, the project did not stop at functional coding, but was carried forward to a physically implementable digital block suitable for higher-level integration.”[1]

#### ### Slide 30 — Conclusion and future work

“To conclude, this work developed and verified an RTL-based digital control architecture for the ROIC, supporting timing control, programmable readout order, windowing, and both IWR and ITR operating modes. In addition, the design was taken through the Cadence RTL-to-GDSII flow, with timing constraints satisfied and implementation feasibility confirmed.”[1]

“As future work, this can be extended toward full mixed-signal top-level integration with the analog ROIC blocks, along with pad ring completion, top-level pin planning, and eventual preparation of a fully integrated tapeout-ready design.”[1]

#### ### Slide 31 — References

“These are the major references used for understanding infrared focal plane arrays, ROIC techniques, detector operation, and prior readout architectures.”[1]

#### ### Slide 32 — Thank you

“Thank you for your time.”[1]

## Better version of your intro

Your intro idea is actually good, but I’d polish it like this for speaking:

### Intro slide 1 polished

“In my previous projects, I had worked with camera signals, image data, and speech-related DSP, but infrared imaging was a new domain for me. So I began from the fundamentals. Infrared lies just beyond visible red in the electromagnetic spectrum, meaning it has longer wavelength and lower frequency than visible light. Since every object above absolute zero emits infrared radiation, IR sensing is strongly connected with temperature and thermal behavior.”[1]

### Intro slide 2 polished

“While studying infrared sensing, I also looked at the physical and material parameters that affect detector behavior. Fundamental laws like Planck’s law, Stefan–Boltzmann law, and Wien’s displacement law help relate wavelength, radiated power, and temperature.”[1]

“On the detector side, material properties such as bandgap, emissivity, and sensitivity are important. HgCdTe is especially useful because its bandgap is tunable across different infrared bands, while InSb is widely used for MWIR detection.”[2][1]

## Small edits I strongly recommend

- Change **“below visible light in terms of freq or energy”** to **“beyond visible red, with longer wavelength and lower frequency/energy than visible light.”**[1]

- Change **“InSb has sensitive for lwir”** to **“InSb is widely used for MWIR detection.”**[2][1]

- On Slide 10, say **“RTL design and simulation in Vivado, synthesis in Genus, implementation in Innovus”** clearly; that line sounds cleaner and more professional.

[1]

- On Slide 30, avoid overstating fabrication unless you truly reached full top-level tapeout readiness; your current future-work slide already frames that correctly.[1]

## ## Presentation tips

- Spend **less time on Slides 3–5** and more time on **Slides 14–30**, because that is your real contribution.[1]
- On waveform slides, don't read every signal; explain the **purpose of the waveform** and what it proves.[1]
- On physical design slides, focus on **flow understanding and outcome**, not tool screenshots alone.[1]

If you want, I can next turn this into:

- a **very natural Hindi-English viva style speech**, or
- a **short 5-minute version** and **full 10-minute version**.